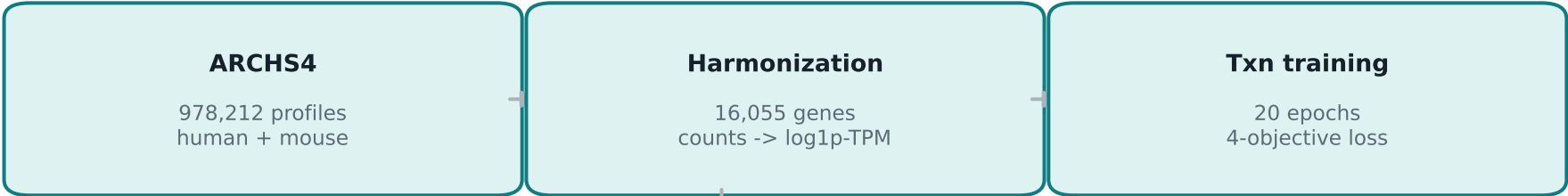


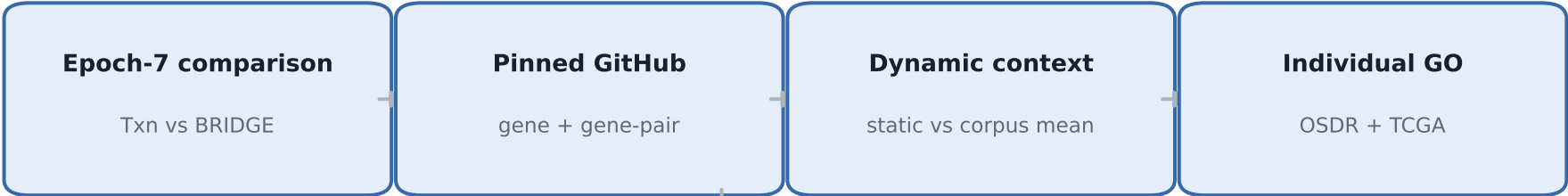
D End-to-end experimental pipeline

From harmonized pretraining to external reconstruction, biological probing, and clinical transfer

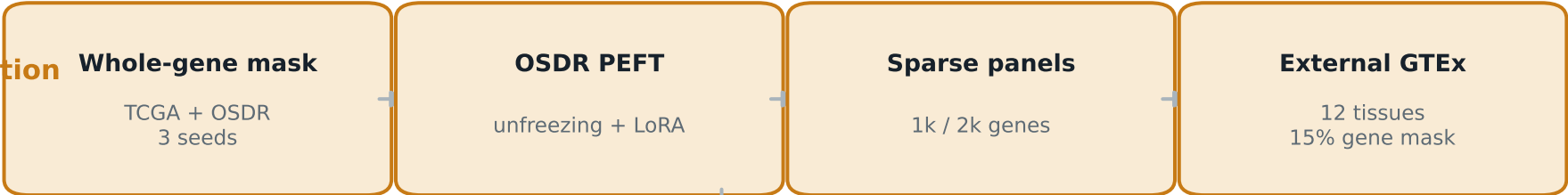
1. Data and pretraining



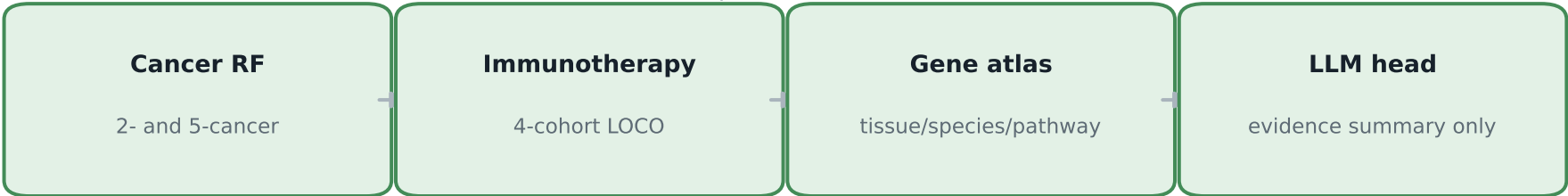
2. Representation audit



3. Reconstruction and adaptation



4. Downstream translation



Validation principle fixed intersections | grouped or cohort-held-out splits | unsupported outputs = NaN | measured results only

Chronology follows the repository experiment order; exploratory and confirmatory analyses remain explicitly separated.